

Syllabus for ECE 602, Lumped Linear Systems (On-Campus Section)

Fall 2026

Course Information

- **Time and Location:** TTh 12-1:15pm at BHEE 222
- **Course credit hours:** 3
- **Prerequisites:**
 - It is expected that you are familiar with the Laplace transform and ordinary differential equations. Knowledge of undergraduate feedback control (at the level of ECE 382) is not strictly needed for most topics; however, it will benefit the understanding of some concepts to be covered by this course (e.g. state space realization, BIBO stability).
 - Knowledge of linear algebra is **needed**. Although officially Purdue's MATH 511 is listed as a co-prerequisite, in practice, students who took MATH 511 and this course in the same semester may find it challenging as the pace of the two courses may not be fully synchronized. Some references that can help you freshen up on linear algebra include:
 - *Linear Algebra and Its Applications*, 4th ed., G. Strang, 2006.
 - *Introduction to Linear Algebra*, 4th ed., G. Strang, Wellesley-Cambridge Press, 2009.

Instructor and TA Contact Information

- **Instructor:** Jianghai Hu, jianghai@purdue.edu
 - **Office hours:** Tuesday 4-5pm EST via Zoom at <https://purdue-edu.zoom.us/j/2693447740>
- **TA:** Somtochukwu (Somto) Marcellinu Oguchienti, soguchie@purdue.edu
 - **Office hours:** Wednesday 11am-12pm EST via Zoom at <https://purdue-edu.zoom.us/j/8316674500>
- Office hours may change in the semester. Consult the Brightspace page for up-to-date information. For in-person meetings, email for appointments.

Online Discussion Forum

A Piazza forum for this course has been set up. You can sign up using the link in Brightspace or directly at:

<https://piazza.com/purdue/fall2026/ece602oncampus>

You are encouraged to post questions and start discussions on anything related to the course on the forum. You can use latex for formulas and post captured pictures/images. If you prefer, your posts can be anonymous or private to the instructor only. While the instruction team will try to respond to your questions timely, other students are welcome to share their thoughts.

Note that there is a separate Piazza forum for the online section of ECE 602. Please **do not** use that forum as the contents of the two sections are different.

Course Description

This course introduces the fundamentals of modern control theory for linear dynamical systems. The course adopts the state-space method that goes beyond the classical transfer function methods covered in undergraduate feedback control courses. The state-space framework is used in modeling, analysis, and controller design for systems arising in many engineering and non-engineering disciplines.

Learning Outcomes

After completing this course, you will be able to:

- Construct models for dynamical systems arising in various applications, such as: mechanical, electrical, pneumatic, hydraulic, economic, and biological systems
- Recognize various properties of given linear systems, such as: stability, controllability, observability, stabilizability, and detectability
- Design controllers so that a system satisfies given performance specifications
- Test and validate the controller design using simulation tools, such as MATLAB and Simulink

Learning Resources, Technology & Texts

Textbook

No required textbooks for this course. The lectures are mostly based on instructor's own notes.

However, the following may be useful references:

- C.-T. Chen, "Linear System Theory and Design," Oxford University Press, Fourth edition, 2013
- S. H. Zak, "[Systems and Control](#)," Oxford University Press, 2003
- P. J. Antsaklis and A. N. Michel, "[A Linear Systems Primer](#)," Birkhauser Boston, 2007
- T. Kailath, "Linear Systems," Prentice Hall, 1980
- W. J. Rugh, "Linear System Theory," Pearson, Second edition, 1995

Linked titles above are available online through the [Purdue Libraries](#).

Required Computational Tools

You will need access to **MATLAB**® for homework. A student edition is sufficient. You can access MATLAB through the [Purdue University MATLAB Portal](#).

Assignments and Grading

Assessment Type	Description	% of Final Grade
Homework	Homework assignments will be given roughly bi-weekly. Submission of the homework will be via Gradescope	25%
Midterms 1&2	Tentatively scheduled to be on Oct. 8 and Nov. 19 (in-class)	40% (20% each)
Final Exam	The final exam will be comprehensive. Date and location TBA	35%

For the exams you are only allowed to bring a non-graphing calculator and a letter-size crib sheet (single-side for the midterms and double-side for the final). The midterms will be in class, and the final's date and location will be available about one month after the semester starts.

The final grades will be determined based on the above weights on a curve with cutoff points decided after the final exam scores become available. The cutoff points will not be made public.

Lecture Schedule

Topics	Number of lectures (estimated)
Systems and state variables	1
State-space models of lumped linear systems	2
Linear algebra review	1
Functions of square matrices	1
Matrix exponential	2
Solution of linear time-invariant systems	2
Solution of linear time-varying systems	1
Stability	2
Lumped nonlinear systems	1
Quadratic forms and singular value decomposition	1
Controllability	2
Observability	2
Minimality, BIBO stability and canonical forms	1
State feedback control	2
Output feedback observer	2
Linear quadratic regulation	3
Model order reduction	1

Note that the above lecture numbers are just estimates. The actual numbers and some of the topics may be adjusted during the semester.

Academic Integrity

Academic integrity is one of the highest values that Purdue University holds. Individuals are encouraged to alert university officials to potential breaches of this value by either emailing integrity@purdue.edu or by calling 765-494-8778. While information may be submitted anonymously, the more information is submitted the greater the opportunity for the university to investigate the concern. More details are available on our course Brightspace table of contents, under University Policies.

Emergency Preparation

In the event of a major campus emergency, course requirements, deadlines and grading percentages are subject to changes that may be necessitated by a revised semester calendar or other circumstances beyond the instructor's control. Relevant changes to this course will be posted onto the course website or can be obtained by contacting the instructors or TAs via email or phone. You are expected to read your @purdue.edu email on a frequent basis.

Attendance Policy

This course follows Purdue's academic regulations regarding attendance, which states that students are expected to be present for every meeting of the classes in which they are enrolled. Guidance on class attendance related to COVID-19 are outlined in the [Protect Purdue Pledge](#) on the Protect Purdue website.

Nondiscrimination Statement

Purdue University is committed to maintaining a community which recognizes and values the inherent worth and dignity of every person; fosters tolerance, sensitivity, understanding, and mutual respect among its members; and encourages each individual to strive to reach his or her potential. In pursuit of its goal of academic excellence, the University seeks to develop and nurture diversity. The University believes that diversity among its many members strengthens the institution, stimulates creativity, promotes the exchange of ideas, and enriches campus life.

Accessibility

Purdue University strives to make learning experiences as accessible as possible. If you anticipate or experience physical or academic barriers based on disability, you are welcome to let me know so that we can discuss options. You are also encouraged to contact the Disability Resource Center at: drc@purdue.edu or by phone: 765-494-1247.